

A Conjecture on Digit Sum Iteration and Prime Numbers

Andrija Zdravković

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Abstract

We conjecture that for every integer $n \geq 1$ with $3 \nmid n$, the sequence defined by repeatedly adding the digit sum of the current term eventually reaches a prime number. We present computational verification up to 10^{12} and discuss why multiples of 3 are necessarily excluded.

1 Introduction

Let $S(n)$ denote the sum of the decimal digits of n .

Define the sequence $(a_k)_{k \geq 0}$ by

$$a_0 = n, \quad a_{k+1} = a_k + S(a_k).$$

We ask: for which starting values n does this sequence eventually contain a prime?

2 The Conjecture

Conjecture (2026). *For every positive integer n not divisible by 3, the sequence (a_k) eventually reaches a prime. Formally:*

$$\forall n \in \mathbb{N}, 3 \nmid n \implies \exists k \in \mathbb{N}_0 : a_k \in \mathbb{P}$$

where $\mathbb{P}$ denotes the set of prime numbers, $a_0 = n$, and $a_{k+1} = a_k + S(a_k)$.

3 Exclusion of Multiples of 3

The condition $3 \nmid n$, for $n > 3$ is necessary. It is well known that $3 \mid n \iff 3 \mid S(n)$. Therefore, if $3 \mid n$ and $n > 3$, then

$$3 \mid (n + S(n)) = a_1,$$

and by induction $3 \mid a_k$ for all $k \geq 0$. Since the sequence is strictly increasing and no multiple of 3 greater than 3 itself is prime, the sequence never reaches a prime. The condition $3 \nmid n$, for $n > 3$ is therefore both necessary and natural.

4 Computational Verification

To extend the verification to large bounds, the iteration process was implemented in C using a bit-packed segmented sieve of Eratosthenes.

We verified the conjecture for all integers $1 \leq n \leq 10^{12}$ with $3 \nmid n$. No counterexample was found. The maximum number of steps required to reach a prime was 430, achieved at $n = 994,370,355,961$.

5 Conclusion

We have stated a new conjecture on digit sum iteration and provided computational evidence up to 10^{12} , with no counterexample found. A proof remains open, and we welcome further work on the problem.

6 Appendix

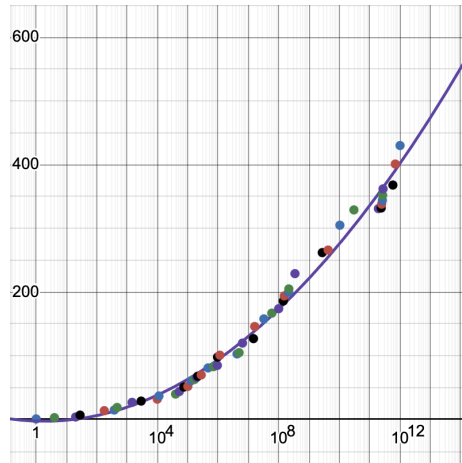


Figure 1: Maximum step count observed as a function of n , plotted on a logarithmic n -axis, with best-fit curve $f(n) = 0.554(\ln n)^2 - 0.668 \ln n - 2.470$ ($R^2 \approx 0.989$).

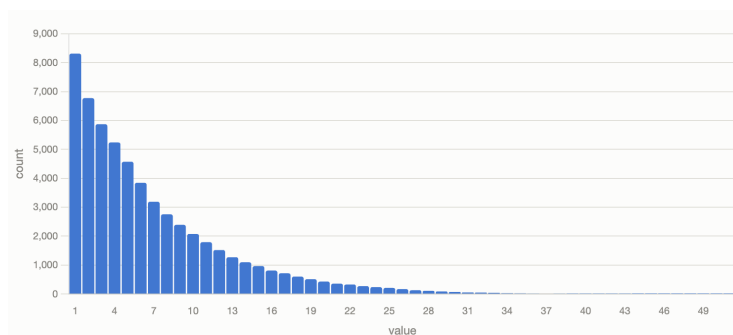


Figure 2: Distribution of step counts, for $n = 1$ to 100,000.

The distribution decays approximately exponentially in step count, consistent with a roughly constant per-step chance of hitting a prime.